

HOBART, OK, USA

WMO: 723525

Lat: 34.989N

Lon: 99.053W

Elev: 1556

StdP: 13.89

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93986

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	18.9	2.6	6.7	20.4	6.9	8.3	25.6	33.3	45.9	30.5	46.4	11.3	20	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	103.4	72.2	100.8	72.2	98.2	72.1	76.9	91.5	75.6	91.0	74.7	90.1	13.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	130.5	82.2	72.0	125.2	80.9	70.7	119.8	80.0	41.4	91.4	40.2	90.6	39.2	90.0	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.0	26.5	24.3	DB	6.9	107.0	4.5	3.3	3.6	109.4	1.0	111.3	-1.5	113.1	-4.7	115.5	
			WB	6.1	79.8	4.4	3.4	2.9	82.2	0.3	84.1	-2.2	86.0	-5.4	88.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	39.3	43.0	51.4	59.8	70.2	79.8	84.1	83.1	75.2	62.7	50.5	40.4
	DBStd	18.14	9.51	10.85	10.29	8.60	8.32	5.84	5.34	5.56	7.47	8.97	9.48	9.25
	HDD50	1155	348	241	107	19	1	0	0	0	0	12	108	319
	HDD65	3454	796	616	431	198	45	1	0	0	12	150	441	764
	CDD50	5436	17	46	151	313	625	895	1058	1025	757	407	122	20
	CDD65	2259	0	1	10	42	204	446	593	560	318	80	5	0
	CDH74	27475	4	31	123	489	2232	5343	7932	7098	3366	784	70	3
	CDH80	14867	0	6	23	141	1002	2892	4736	4136	1660	259	12	0
Wind	WSAvg	12.4	12.3	12.9	13.6	13.7	12.3	13.0	11.6	10.8	11.5	12.7	12.7	12.0
Precipitation	PrecAvg	26.5	0.7	1.0	1.7	2.1	4.4	3.3	2.3	2.6	3.3	2.7	1.5	1.1
	PrecMax	36.2	3.0	3.0	5.1	7.0	12.0	9.7	10.6	7.5	10.9	10.4	6.7	4.4
	PrecMin	13.3	0.0	0.0	0.0	0.0	0.1	0.6	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.9	0.8	0.9	1.2	1.5	3.2	2.4	1.9	2.0	2.8	2.2	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.6	78.3	83.3	89.8	99.2	105.0	106.8	106.5	102.4	91.3	80.9	70.3
		MCWB	53.3	55.9	59.8	63.9	69.5	71.3	72.0	71.2	72.5	67.9	61.9	53.7
	2%	DB	65.9	71.8	77.3	84.0	94.3	99.9	103.5	103.0	97.2	86.8	74.9	64.3
		MCWB	49.9	54.8	59.0	64.0	69.5	71.7	72.0	72.4	71.1	66.5	59.6	53.3
	5%	DB	60.8	65.7	72.9	79.7	90.2	96.7	101.0	100.2	93.4	82.3	70.2	59.9
		MCWB	47.8	52.7	59.0	62.7	68.8	72.2	72.5	72.4	70.7	65.1	57.4	49.5
	10%	DB	55.6	60.8	68.6	75.5	85.5	93.5	98.2	97.1	89.7	78.2	65.6	55.3
		MCWB	45.3	50.5	57.0	61.8	68.0	72.3	72.7	72.3	69.7	63.0	56.0	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	62.1	67.5	71.7	75.2	78.9	78.6	77.5	77.3	73.1	66.7	61.0
		MCDB	65.3	69.4	75.7	79.7	88.4	92.1	92.3	91.5	93.0	83.4	73.2	64.7
	2%	WB	52.9	58.3	64.4	68.7	73.3	76.5	76.7	76.1	74.6	70.6	62.6	55.7
		MCDB	62.0	66.5	71.9	77.0	86.2	90.9	91.8	91.5	88.2	79.4	70.4	61.0
	5%	WB	49.1	54.8	61.8	66.6	71.7	75.2	75.4	75.0	73.2	68.4	60.1	51.2
		MCDB	59.2	64.0	68.7	74.8	84.0	90.1	91.4	90.9	86.2	76.6	67.3	58.0
	10%	WB	45.6	50.9	58.6	64.3	70.3	73.9	74.4	74.0	71.8	66.1	57.4	46.9
		MCDB	55.3	60.0	66.9	72.1	82.1	88.4	90.6	89.8	84.1	75.0	64.3	54.6
Mean Daily Temperature Range	5% DB	MDBR	23.2	23.5	24.0	24.5	23.8	23.4	24.0	23.6	22.8	23.2	22.6	21.1
		MCDBR	32.4	32.1	29.6	30.4	28.9	27.4	27.6	27.6	26.9	27.8	28.5	28.4
		MCWBR	19.7	18.9	16.7	15.5	10.7	8.4	6.9	7.9	9.1	12.5	15.8	17.7
	5% WB	MCDBR	29.4	27.7	23.6	23.5	24.5	23.2	23.6	23.3	22.0	21.1	23.4	24.4
		MCWBR	19.3	18.8	15.2	15.0	11.5	9.1	7.7	7.7	8.8	11.8	14.9	17.2
Clear-Sky Solar Irradiance	taub	0.282	0.294	0.329	0.363	0.388	0.413	0.413	0.407	0.374	0.344	0.308	0.285	
	taud	2.529	2.503	2.421	2.354	2.356	2.324	2.355	2.319	2.388	2.459	2.494	2.519	
	Ebn at Noon	297	304	299	292	285	276	275	275	280	281	283	288	
	Edh at Noon	25	29	35	39	39	41	39	40	36	30	26	24	
All-Sky Solar Radiation	RadAvg	948	1146	1519	1861	2012	2225	2219	2015	1685	1283	989	805	
	RadStd	92	152	100	143	210	187	123	118	114	171	124	81	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page